The Limits of the Expectations Hypothesis

Supplementary Material

This supplement documents the forecasting protocol and the evidence behind the three results of the letter. All tables come from the same programs, data, and samples as the letter. The research uses the overnight, one-week, two-week, and one-month repo rates (pairs ON-1W, ON-2W, ON-1M, and 1W-2W), the four Bloomberg composites that are updated throughout the sample; the 3W, 2M, and 3M composites have not changed since December 2015. The stale-quote filter for the four maturities is unaffected by the exclusion of the other three. The excess return of rolling the short repo over holding the long repo is

$$xr_t^{(n,m)} = \left(r_t - r_t^f\right)\frac{360}{D_t}, \qquad r_t^f = i_t^{(n)}\frac{D_t}{360}, \qquad r_t = \prod_{j=0}^{k-1}\left(1 + i_{t_j}^{(m)}\frac{d_j}{360}\right) - 1, \tag{S1}$$

and the expectations hypothesis (EH) with a constant premium implies $E_t[xr_t^{(n,m)}] = -\pi^{(n,m)}$, so that the prevailing mean is the EH forecast. Section S2 refers to the VAR restrictions of Della Corte, Sarno, and Thornton (2008) (DST),

$$e_2' = e_1'\frac{1}{k}\sum_{j=0}^{k-1} G^{mj}, \tag{S2}$$

where G is the companion matrix of a bivariate VAR(5) of the short and the long rate and e_1, e_2 select the two rates.

S1 Forecasting Protocol and Models

Table S1 lists the elements of the forecasting design, which are identical for every model. Table S2 summarises the model classes and the departures from the EH they can represent, and Table S3 the hyperparameter grids.

Table S1: **Forecasting Protocol: Identical for Every Model**

Element	Specification (all models)
Target	Annualised excess return of rolling the short over holding the long repo, Equation (S1)
Benchmark	EH forecast: prevailing mean of the excess returns in the estimation window
Predictors	Five daily lags of the two rates of the pair (DST's VAR(5) information set); same for every model
Forecast origin	Every business day t that is not a US holiday; information dated t or earlier
Estimation window	Rolling 1,250 business days; only pairs (x_s, xr_s) with $s + n \leq t$, so no target is realised after t
Re-estimation	Every 21 business days; hyperparameters re-tuned every 252 business days
Tuning	Last 15% of the estimation window, separated from the training targets by $n - 1$ observations; objective: mean squared error; grids in Table S3
Preprocessing	Standardisation with the mean and standard deviation of the estimation window; winsorisation at ± 5; principal components (PCR) estimated inside the window
Missing data	Stale quotes (> 20 identical business days) and one typo set to missing; no imputation; a forecast requires all predictors
Stochastic training	Each network forecast averages three initialisations; the ensemble of five architectures and the equal-weighted combination of all forecasts are fixed ex ante
Evaluation	R^2_{oos} on common dates; Clark–West and Diebold–Mariano tests with Newey–West variances (at least n lags); SPA, model confidence set, and Holm adjustment across all forecasts

Tuning budgets differ across classes, from no tuning (VAR, bagging) to 40 configurations for the networks (eight per architecture); Table S3 lists the grids. The design was fixed on data to April 2021 and applied unchanged to the holdout from 21 April 2021.

Table S2: **Model Classes and the Departures from the EH They Can Detect**

Class	Methods	Departures they can detect	Main limitation
EH forecast	Prevailing mean	None (constant premium)	–
VAR benchmark	OLS on lags of the two rates	Linear functions of recent rates, as in DST	Unstable with many correlated regressors
Regularised linear	Ridge, lasso, elastic net, PCR, PLS, linear SVR	Linear functions; robust to collinearity and noise	No nonlinearities or interactions
Kernel and local	RBF SVR, nearest neighbours	Smooth nonlinear relations, local patterns	Curse of dimensionality
Tree-based	Tree, random forest, boosting, bagging	Thresholds, regimes, interactions	Piecewise constant, poor extrapolation
Neural networks	1 to 5 hidden layers, ensemble	Smooth nonlinearities and interactions of any form	Need large signal-to-noise and regularisation
Reference forecasts	Combination of all forecasts; EWMA of past excess returns	Robust aggregation; slowly moving premium	No new information

All models use the same predictors, estimation windows, re-estimation dates, and hyperparameter tuning on the same validation block (Section S1). Under the EH, none of them can forecast the excess return of rolling over holding better than the prevailing mean. Hyperparameter grids are in Table S3 of the Online Appendix.

Table S3: **Models and Hyperparameters**

Model	Hyperparameters
OLS / VAR	not tuned
Ridge	`alpha` $\in \{10^{-2}, 10^{-1.5}, \ldots, 10^{3}\}$ (11 points)
Lasso	`alpha` $\in \{10^{-4}, 10^{-3.5}, \ldots, 10^{0}\}$ (9 points)
Elastic net	`alpha` $\in \{10^{-4}, 10^{-3.5}, \ldots, 10^{0}\}$ (9 points); `l1_ratio` $\in$ {0.1, 0.5, 0.9}
PCR	`n_components` $\in$ {1, 2, 3, 5, 8}
PLS	`n_components` $\in$ {1, 2, 3, 5}
Linear SVR	`C` $\in \{10^{-3}, 10^{-2}, \ldots, 10^{1}\}$ (5 points); `epsilon` $\in$ {0, 0.1}
RBF SVR	`C` $\in$ {0.1, 1, 10}; `gamma` $\in$ {scale, 0.01}; `epsilon` $\in$ {0.05, 0.2}
KNN	`n_neighbors` $\in$ {5, 10, 20}; `weights` $\in$ {uniform, distance}
Tree	`max_depth` $\in$ {2, 3, 5}; `min_samples_leaf` $\in$ {5}
Random forest	`max_depth` $\in$ {3, 6, ∞}; `max_features` $\in$ {0.33, 1}; `min_samples_leaf` $\in$ {5}; fixed: `n_estimators`=300
Boosting	`n_estimators` $\in$ {100, 300}; `learning_rate` $\in$ {0.01, 0.1}; `max_depth` $\in$ {1, 2, 3}; fixed: `subsample`=0.8
Bagging	not tuned; fixed: `n_estimators`=100
Random features (ridge)	`n_features` $\in$ {100, 1e+03, 5e+03}; `gamma` $\in$ {0.02, 0.1}; `alpha` $\in$ {0.1, 10, 1e+03}
Neural networks	architectures (2), (4-2), (8-4-2), (16-8-4-2), (32-16-8-4-2); `dropout` $\in$ {0, 0.2}; `l1l2` $\in$ {0.0001, 0.001}; `learning_rate` $\in$ {0.001, 0.01}; Adam, batch 256, early stopping with patience 10, at most 200 epochs, 3 seeds

Forecasting models and hyperparameter grids (daily data). Each model is estimated on predictors and a target standardised with the mean and standard deviation of the estimation window; standardised predictors are winsorised at ± 5. Hyperparameters are chosen on the last 15% of the estimation window by minimising the mean squared forecast error. Each hidden layer of a network applies a dense layer with an elastic-net penalty, batch normalisation, a ReLU activation and dropout; forecasts average the networks of the different seeds. The table is generated from the code.

S2 In-Sample Tests and Their Size

Table S4 reports DST's Wald test for our sample. Table S5 reports tests for neglected nonlinearity of the in-sample projection and the size of the Wald test in samples simulated from a two-factor model in which the EH holds exactly. The model is stylised: several persistence estimates lie at the upper bound of 0.9999, and for some pairs the simulated rates range more widely than the data. The size result should therefore be read as evidence that the asymptotic critical values are unreliable for near-integrated daily rates with overlapping returns, not as an exact size.

Table S4: **In-Sample VAR Tests of the Expectations Hypothesis**

$i^{(m)}$–$i^{(n)}$	k	Wald	p-value	$\rho(i^{(n)}, i^{(n)*})$	$\sigma(i^{(n)*})/\sigma(i^{(n)})$	Obs.
ON-1W	5	452.8	0.0000	0.9993	0.9991	5650
ON-2W	10	242.1	0.0000	0.9998	0.9975	4216
ON-1M	21	3132.4	0.0000	0.9982	1.0020	4059
1W-2W	2	3084.3	0.0000	0.9995	0.9994	4276

Wald test of the nonlinear cross-equation restrictions that the expectations hypothesis imposes on a bivariate VAR(5) of the short rate $i^{(m)}$ and the long rate $i^{(n)}$, $e_2' = e_1' k^{-1} \sum_{j=0}^{k-1} G^{mj}$ (Equation S2). The covariance matrix of the VAR coefficients is heteroskedasticity robust; the statistic is $\chi^2(10)$ under the null. $i^{(n)*}$ is the theoretical long rate implied by the EH and the VAR; the last columns report its correlation with and relative volatility to the actual long rate. Daily data, stale quotes removed.

Table S5: **Preliminary Evidence: Misspecification and Size of In-Sample Tests**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: In-sample projection on the VAR information set				
R^2 (%)	44.7	23.5	11.0	48.3
Wald test, p-value	0.000	0.000	0.000	0.000
Panel B: Tests for neglected nonlinearity				
Neural-network test, median p-value	0.697	0.080	0.422	0.606
Neural-network test, Bonferroni p-value	0.787	0.024	1.000	0.932
RESET test, p-value	0.701	0.001	0.595	0.011
Panel C: Wald test in 200 samples simulated under the EH				
Persistence of the short rate φ_x	0.5637	0.9999	0.7582	0.9999
Persistence of investors' information φ_z	0.9997	0.6534	0.9999	0.8409
Rejection rate at the 5% level (%)	87.5	96.5	97.0	97.0
95th percentile of the Wald statistic	$\gg 0$	$\gg 0$	$\gg 0$	$\gg 0$
Wald statistic in the data	452.8	242.1	3132.4	3084.3
Bootstrap p-value	0.535	0.845	0.325	0.840
Obs.	5643	4185	3987	4261

Panel A: in-sample regression of the excess return of rolling the short repo over holding the long repo on a constant and five daily lags of the two rates (the information set of the VAR of DST), and the p-value of the Wald test that all slopes are zero. Panel B: tests of the null that this linear projection is correctly specified. The neural-network test of Lee, White, and Granger (1993) regresses the residuals on the regressors and the second to fourth principal components of the activations of 20 logistic hidden units with random weights; it is repeated for ten draws of the weights, and the Bonferroni p-value is ten times the smallest p-value. The RESET test of Ramsey (1969) adds the squared and cubed fitted values. All tests use Newey–West covariances with the automatic bandwidth of Andrews (1991), at least the forecast horizon. Panel C: rates simulated from a two-factor model in which the EH holds exactly, $x_{t+1} = \varphi_x x_t + z_t + \varepsilon_{t+1}$, $z_{t+1} = \varphi_z z_t + \eta_{t+1}$, with x_t the deviation of the short rate from its mean and the long rate equal to the average expected short rate plus a constant; φ_x and φ_z are estimated by Gaussian maximum likelihood, the shocks are drawn by a wild bootstrap of the estimated shocks, and the simulated rates have the stale-quote pattern of the data. The Wald test of Table S4 is computed on each sample; the bootstrap p-value is the share of simulated statistics at least as large as in the data. Daily data, 23 July 1998 to 30 September 2026.

S3 All Forecasts and Joint Tests

Table S6 reports the out-of-sample R^2 of all forecasts, including a pooled network estimated on the stacked observations of all pairs. Table S7 reports the tests across models (SPA, model confidence set, Holm adjustment, encompassing, direction, and conditional predictive ability). Table S8 compares each learner with the VAR, and Table S9 shows how far each learner's forecasts depart from the EH forecast relative to the VAR. Figure S1 shows when the gains accrue.

Table S6: **Out-of-Sample Predictability of Departures from the EH**

	ON-1W	ON-2W	ON-1M	1W-2W
VAR (DST)	38.68***	25.40***	−17.26	49.55***
Ridge	21.32***	22.33***	−18.01	48.25***
Lasso	21.74***	12.38***	−22.11	41.76***
Elastic net	21.33***	12.54***	−22.53	41.86***
PCR	20.28***	22.70***	−13.04	44.99***
PLS	19.03***	20.39***	−18.07	45.61***
Linear SVR	20.99***	26.27***	−1.04*	47.94***
RBF SVR	15.63***	20.83***	−66.48	32.24***
KNN	−0.73***	14.29***	−17.59**	29.39***
Tree	−72.61	−18.48***	−86.43**	13.05***
Random forest	−54.38	−1.43***	−56.43*	27.34***
Boosting	−72.97*	6.24***	−34.37*	26.52***
Bagging	−66.09**	−18.93***	−38.89**	15.73***
Random features	1.34***	17.90***	−54.27	11.05***
NN (2)	3.95***	6.81***	−14.88	8.55***
NN (4-2)	−0.22	−1.28*	−11.18	11.65***
NN (8-4-2)	−2.59	−4.24	−7.30	−2.11
NN (16-8-4-2)	−2.55	−1.33**	−5.69	−2.73
NN (32-16-8-4-2)	2.38**	0.60**	−4.36	1.29***
NN ensemble	1.89***	3.38***	−4.15	5.93***
Combination	16.16***	23.32***	−4.78*	40.71***
EWMA mean	0.69***	8.68***	−2.24**	18.82***
Pure EH (zero)	−4.38	−10.79	−5.87	17.51***
Spread only	34.63***	25.84***	0.24**	47.21***
Pooled ridge	22.55***	26.34***	−5.99***	33.82***
Pooled NN ensemble	4.85***	4.10***	−4.49	5.18**
Obs.	4666	3277	3088	3332

R^2_{oos} (in percent) of forecasts of the annualised excess return of rolling the short repo $k = n/m$ times over holding the long repo, relative to the prevailing mean, which is the forecast implied by the expectations hypothesis with a constant term premium. Stars: test of Clark and West (2007) of $H_0 : R^2_{oos} \leq 0$ at the 10%, 5% and 1% level (Newey–West variance with the automatic bandwidth of Andrews (1991), at least the horizon). Forecasts are made every business day from rolling five-year windows; models are re-estimated every 21 business days and re-tuned every 252. Within a table all models are evaluated on the same dates. 'Pure EH (zero)' is the zero forecast, i.e. the EH with a zero term premium. Predictors: five lags of the two rates of each pair (the VAR information set). Pooled: one model estimated on the stacked, pair-wise standardised observations of all pairs with the maturities as additional inputs (Section S3); the network ensemble averages the 8-4-2 and 32-16-8-4-2 architectures; evaluated on the dates on which both the pooled and the pair forecasts exist, and not reported (–) for fewer than 200 dates. Spread only: regression on the spread between the long and the short rate at t, the predictor of the classical regression tests, evaluated on the dates on which both the spread and the pair forecasts are available.

Table S7: **Joint Tests across Models**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: Does any model beat the EH?				
SPA p-value (best model vs EH)	0.011	0.000	1.000	0.010
MCS p-value of the EH forecast	0.372	0.185	1.000	0.123
MCS p-value of the VAR	1.000	0.729	0.849	1.000
Models in the 90% MCS	23/23	20/23	23/23	21/23
Clark–West rejections, unadjusted	16/22	20/22	4/22	20/22
Clark–West rejections, Holm within pair	13/22	18/22	0/22	20/22
Clark–West rejections, Holm across pairs	11/22	15/22	0/22	10/22
Panel B: Does machine learning add to the VAR?				
Learners not encompassed by the VAR	0/19	8/19	13/19	7/19
after Holm adjustment	0/19	5/19	0/19	1/19
Panel C: Direction and state dependence				
Hit rate of the VAR (%)	71.9***	75.0***	54.7	77.5***
Hit rate of the NN ensemble (%)	58.2***	58.8***	50.7	60.8**
Giacomini–White p-value (VAR vs EH)	0.005	0.000	0.240	0.000
Obs.	4666	3277	3088	3332

Panel A: p-value of the test of superior predictive ability of Hansen (2005) of the null that no model forecasts the excess return better than the EH forecast (prevailing mean); p-values of the EH forecast and of the VAR in the model confidence set of Hansen, Lunde, and Nason (2011) (a forecast is in the 90% set if its p-value is at least 0.10) and the size of the set; and the number of models for which the Clark–West test rejects at the 5% level, without adjustment and with the Holm (1979) adjustment within each pair and across all pairs. The model set comprises the VAR, 18 learners (13 methods and five network architectures), the network ensemble, the equal-weighted combination of all forecasts, and an exponentially weighted mean of past excess returns; losses are squared errors, and the bootstraps use the stationary bootstrap of Politis and Romano (1994) with a mean block length of 20 days or twice the forecast horizon, whichever is larger. Panel B: number of learners (including the network ensemble) for which the encompassing test of Harvey, Leybourne, and Newbold (1998) rejects the null that the VAR encompasses the learner at the 5% level. Panel C: share of non-overlapping forecasts that predict the sign of the departure from the prevailing mean correctly, with stars for the test of Pesaran and Timmermann (1992), and the p-value of the conditional predictive ability test of Giacomini and White (2006) with a constant and a month-turn dummy as instruments. Lags of the pair as predictors.

Table S8: **Machine Learning versus the VAR Benchmark**

	ON-1W	ON-2W	ON-1M	1W-2W
Ridge	−1.28	−2.23**	−0.20	−0.62
Lasso	−1.24	−4.55***	−0.51	−1.49
Elastic net	−1.27	−4.42***	−0.57	−1.47
PCR	−1.34	−3.08***	1.37	−5.60***
PLS	−1.57	−4.20***	−0.15	−4.56***
Linear SVR	−1.39	0.41	1.62	−0.51
RBF SVR	−1.92*	−1.21	−1.58	−2.43**
KNN	−2.75***	−1.61	−0.02	−3.33***
Tree	−1.66*	−2.36**	−2.15**	−3.71***
Random forest	−1.56	−2.28**	−1.83*	−3.33***
Boosting	−1.52	−1.65*	−0.86	−3.15***
Bagging	−1.58	−1.46	−0.96	−3.95***
Random features	−1.55	−1.39	−1.60	−3.11***
NN (2)	−2.68***	−3.34***	0.17	−4.08***
NN (4-2)	−2.89***	−4.31***	0.41	−4.58***
NN (8-4-2)	−2.57**	−4.49***	0.67	−4.46***
NN (16-8-4-2)	−2.90***	−4.24***	0.85	−4.37***
NN (32-16-8-4-2)	−2.65***	−3.83***	0.89	−3.97***
NN ensemble	−2.62***	−3.74***	0.95	−4.10***
Combination	−1.51	−0.57	1.23	−2.17**
EWMA mean	−2.77***	−2.47**	0.85	−3.95***

Statistics of the test of Diebold and Mariano (1995) (Newey–West variance with the automatic bandwidth of Andrews (1991), at least the horizon) comparing each model with the VAR benchmark of Della Corte et al. (2008), the direct linear forecast from five lags of the two repo rates. Positive values indicate a lower mean squared forecast error than the VAR. Stars: two-sided significance at the 10%, 5% and 1% level. All models use the lags of the pair as predictors.

Table S9: **How the Learners Differ from the VAR**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: Dispersion of the departures from the EH forecast relative to the VAR				
Ridge	0.88	0.89	0.90	0.94
Lasso	0.88	0.77	0.82	0.84
Elastic net	0.87	0.77	0.80	0.83
PCR	0.88	0.94	0.95	0.97
PLS	0.85	0.87	0.91	0.96
Linear SVR	0.94	0.98	0.61	0.95
RBF SVR	0.74	0.64	1.50	1.00
KNN	0.59	1.10	1.39	0.95
Tree	1.34	1.48	1.98	1.19
Random forest	1.17	1.40	1.66	1.05
Boosting	1.44	1.23	1.34	1.10
Bagging	1.40	1.84	1.83	1.25
Random features	0.79	0.71	1.27	0.96
NN (2)	0.33	0.55	0.69	0.23
NN (4-2)	0.25	0.51	0.36	0.43
NN (8-4-2)	0.26	0.46	0.72	0.30
NN (16-8-4-2)	0.23	0.63	0.36	0.30
NN (32-16-8-4-2)	0.12	0.16	0.23	0.09
NN ensemble	0.16	0.35	0.38	0.20
Combination	0.56	0.62	0.72	0.67
EWMA mean	0.19	0.45	0.73	0.37
Panel B: Correlation of the departures from the EH forecast with those of the VAR				
Ridge	0.91	0.93	0.94	0.98
Lasso	0.91	0.82	0.84	0.86
Elastic net	0.90	0.82	0.84	0.86
PCR	0.88	0.94	0.95	0.97
PLS	0.87	0.89	0.92	0.98
Linear SVR	0.89	0.86	0.82	0.90
RBF SVR	0.64	0.60	0.40	0.67
KNN	0.24	0.37	0.21	0.61
Tree	0.11	0.27	0.28	0.54
Random forest	0.17	0.32	0.29	0.62
Boosting	0.17	0.32	0.29	0.62
Bagging	0.20	0.36	0.20	0.58
Random features	0.54	0.44	0.32	0.55
NN (2)	0.46	0.47	0.33	0.38
NN (4-2)	0.23	0.25	0.05	0.37
NN (8-4-2)	0.13	0.23	0.42	0.14
NN (16-8-4-2)	0.09	0.24	0.38	0.09
NN (32-16-8-4-2)	0.27	0.13	0.22	0.18
NN ensemble	0.35	0.37	0.39	0.33
Combination	0.74	0.78	0.68	0.86
EWMA mean	0.17	0.19	0.13	0.54
Panel C: Distinct hyperparameter configurations selected				
Ridge	8	5	5	6
Lasso	7	5	7	8
Elastic net	10	6	8	8
PCR	4	3	4	2
PLS	4	4	4	3
Linear SVR	7	6	6	5
RBF SVR	9	5	6	7
KNN	5	4	4	5
Tree	3	3	3	3
Random forest	5	6	4	5
Boosting	9	7	7	6
Random features	9	9	7	6
NN (2)	6	5	5	5
NN (4-2)	7	5	5	6
NN (8-4-2)	8	5	6	5
NN (16-8-4-2)	8	6	5	6
NN (32-16-8-4-2)	8	5	6	6
Number of tuning dates	19	12	9	12

Panel A: standard deviation of the departures of each model's forecast from the EH forecast (prevailing mean), relative to that of the VAR; values below one indicate forecasts that are shrunk towards the EH. Panel B: correlation of these departures with those of the VAR. Panel C: number of distinct hyperparameter configurations selected across the tuning dates (every 252 forecast origins). Lags of the pair as predictors.

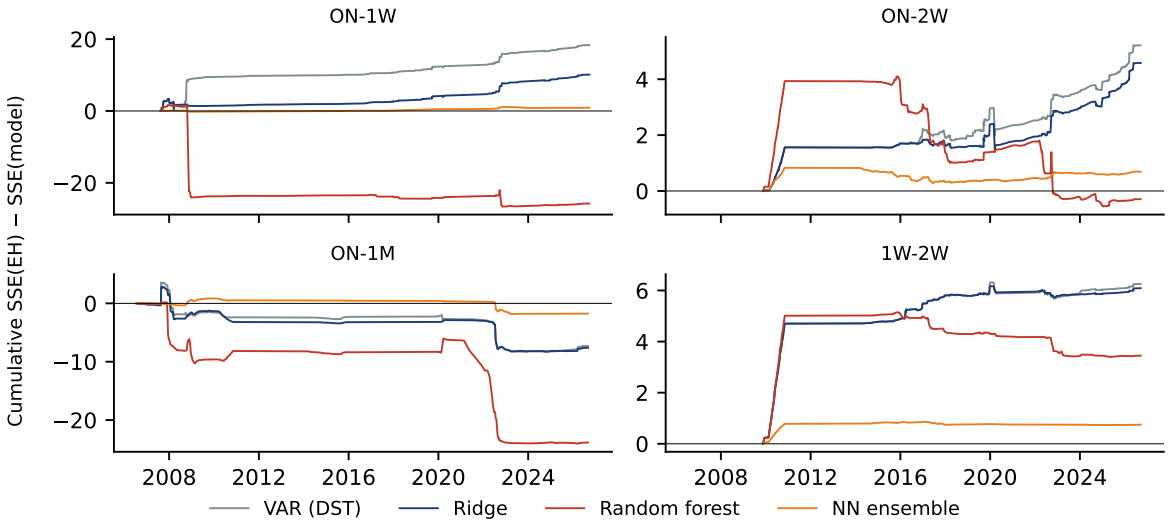


Figure S1: **Cumulative Forecast Gains Relative to the EH.** Cumulative difference between the squared forecast errors of the EH forecast (prevailing mean) and those of the VAR, ridge regression, random forest, and the neural-network ensemble. An increasing line indicates that the model forecasts better than the EH. Predictors: five lags of the two rates of each pair.

S4 Quotes, Month Turns, and a Slowly Moving Premium

Table S10 compares forecasts with and without the quotes of the decision day and uses an exponentially weighted mean of past excess returns as the benchmark. Table S11 compares the composite overnight quote with the transaction-based rates published by the Federal Reserve Bank of New York since April 2018. Table S12 adds calendar dummies to the VAR and to boosted trees and shows where the VAR's gains arise.

Table S10: **Same-Day Quotes and a Slowly Moving Premium**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: R^2_{oos} (%) with today's quotes (lags 0–4) and without them (lags 1–5)				
VAR (DST), lags 0–4	39.09	25.95	−15.84	46.66
VAR (DST), lags 1–5	0.34***	5.12***	−21.11	26.20***
Ridge, lags 0–4	19.89	22.78	−16.58	45.47
Ridge, lags 1–5	0.75***	1.36***	−24.76	26.52***
Boosting, lags 0–4	−77.48	5.47	−34.15	23.03
Boosting, lags 1–5	−73.09	−15.31**	−29.02	−3.11***
Panel B: A slowly moving premium as the benchmark				
R^2_{oos} of the EWMA vs prevailing mean (%)	0.69	8.68	−2.24	18.82
Best model, R^2_{oos} vs EWMA (%)	38.25	19.25	1.17	37.85
SPA p-value (best model vs EWMA)	0.015	0.004	0.884	0.002
Models better than EWMA (Holm)	7/21	2/21	0/21	8/21

Panel A: out-of-sample R^2 relative to the EH forecast of the VAR, ridge regression, and boosted trees with five lags of the two rates dated t to $t-4$ (as in Table S6) and dated $t-1$ to $t-5$, on the same dates; stars refer to the Clark–West test. A transitory error in the quote of the long rate at t enters both the predictors and, with the opposite sign, the excess return; forecasts without today's quotes are free of this channel. Panel B: forecasts compared with an exponentially weighted mean of past excess returns (half-life of six months) instead of the prevailing mean: R^2_{oos} of the EWMA relative to the prevailing mean, the largest R^2_{oos} of any model relative to the EWMA, the p-value of the test of superior predictive ability of Hansen (2005) of the null that no model beats the EWMA, and the number of models that beat it in one-sided Diebold–Mariano tests at the 5% level after the Holm (1979) adjustment.

Table S11: **Composite Quotes versus Transaction-Based Overnight Rates**

	TGCR	BGCR	SOFR
Observations	2120	2120	2120
Mean difference, composite minus reference (bp)	4.68	4.66	2.93
Mean absolute difference (bp)	6.03	6.01	4.49
Share of days with absolute difference above 5 bp (%)	55.9	55.6	32.7
Correlation of levels	0.999	0.999	0.999
Correlation of daily changes	0.397	0.398	0.414
Days without change, composite (%)	10.8	10.8	10.8
Days without change, reference (%)	50.8	50.3	43.6
ON-1W excess return with TGCR as rolled leg: correlation	0.842		

Comparison of the Bloomberg composite overnight GC repo rate with the transaction-based reference rates of the Federal Reserve Bank of New York (tri-party general collateral rate TGCR, broad general collateral rate BGCR, and secured overnight financing rate SOFR), 2 April 2018 to 30 September 2026, excluding holidays. The last row correlates the ON-1W excess return of Equation (S1) with the one in which the rolled overnight leg uses TGCR.

Table S12: **Calendar Effects and the Departures from the EH**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: R^2_{oos} (%)				
VAR (DST)	38.68***	25.40***	−17.26	49.55***
VAR + calendar	37.01***	21.23***	−19.52	48.84***
Calendar only	−2.20	−2.68	0.04**	−0.99**
Boosting	−72.97*	6.24***	−34.37*	26.52***
Boosting + calendar	−63.50*	−1.49***	−37.62*	26.36***
Panel B: Diagnostics				
DM: VAR + calendar vs VAR	−1.61	−2.95***	−1.92*	−1.03
Shapley share of calendar (%)	21.03	20.50	19.60	16.05
Windows with a month turn (%)	22.91	46.05	96.28	45.92
Share of VAR gain in these windows (%)	27.90	76.88	96.91	58.73
R^2_{oos} VAR, windows with a month turn	42.09	33.91	−17.41	54.21
R^2_{oos} VAR, windows without a month turn	37.50	13.85	−13.64	44.16
Obs.	4666	3277	3088	3332

Panel A reports R^2_{oos} relative to the prevailing mean, with Clark–West significance stars, for the VAR benchmark (five lags of the two rates), the VAR augmented with calendar dummies (day of the week; last and first business day of the month, quarter and year; and whether the investment window contains a month, quarter or year turn), a regression on the calendar dummies only, and boosted trees with and without the calendar dummies. All calendar information is known at the forecast origin. Panel B reports a Diebold–Mariano test of the VAR with calendar dummies against the VAR (positive: calendar dummies improve the forecast), the share of the calendar block in the absolute Shapley contributions to the out-of-sample forecasts of the VAR with calendar dummies, the share of forecast origins whose investment window contains a month turn, the share of the VAR's out-of-sample gain over the EH forecast, $\sum_t[(y_t-\bar{y}_t)^2-(y_t-\hat{y}_t)^2]$, earned in these windows, and the VAR's R^2_{oos} in windows with and without a month turn (in percent).

S5 Detectability of Known Departures

Table S13 and Figure S2 report how the models perform on simulated excess returns with a known linear or nonlinear departure from the EH, and Table S14 how often the Clark–West test rejects. Each cell averages three noise draws, so the rejection rates are illustrative rather than precise power estimates. The economic bound of the letter does not rely on these simulations: it is computed for a perfect-foresight investor on the actual data (Table 2 of the letter).

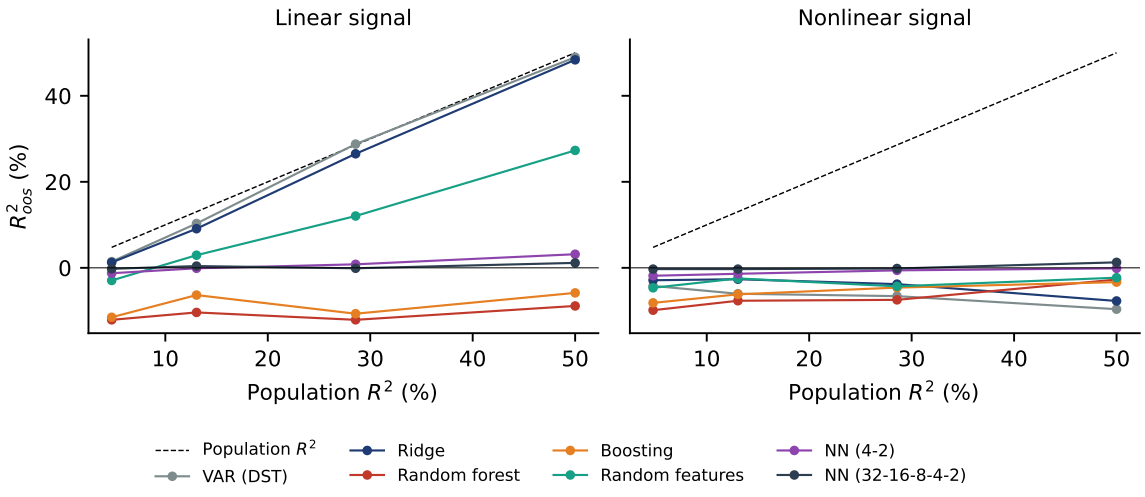


Figure S2: **Recovering a Known Departure from the EH.** Out-of-sample R^2 of the models on simulated excess returns that contain a linear (left) or nonlinear (right) departure from the EH, against the population R^2 of the departure (dashed line). Predictors, sample gaps, and backtest design of the ON-1W pair; averages over three noise draws.

Table S13: **Can the Learners Recover a Known Signal? Signal-to-Noise Simulation**

SNR	Population R^2	Oracle	VAR (DST)	Ridge	Random forest	Boosting	Random features	NN (4-2)	NN (32-16-8-4-2)
Panel A: linear signal									
0.05	4.8	5.2	1.4 (0.8)	1.3 (0.7)	−12.1 (2.2)	−11.5 (5.2)	−2.9 (2.1)	−1.3 (0.2)	−0.2 (0.4)
0.15	13.0	13.9	10.3 (2.0)	9.1 (1.9)	−10.4 (3.7)	−6.3 (3.6)	2.9 (2.4)	−0.1 (0.4)	0.4 (0.4)
0.40	28.6	30.9	28.8 (1.5)	26.5 (0.2)	−12.1 (3.2)	−10.7 (5.3)	12.0 (0.9)	0.8 (0.8)	−0.1 (0.4)
1.00	50.0	51.2	49.0 (1.3)	48.4 (1.6)	−8.9 (4.8)	−5.8 (4.6)	27.3 (2.9)	3.2 (2.9)	1.2 (1.2)
Panel B: nonlinear signal									
0.05	4.8	5.2	−4.2 (1.3)	−2.9 (1.8)	−9.9 (2.1)	−8.1 (1.9)	−4.6 (3.3)	−1.8 (0.6)	−0.3 (0.3)
0.15	13.0	14.5	−6.0 (0.0)	−2.6 (1.6)	−7.7 (0.9)	−6.2 (2.0)	−2.5 (1.6)	−1.4 (0.1)	−0.3 (0.3)
0.40	28.6	31.3	−6.6 (0.6)	−3.8 (2.7)	−7.4 (1.5)	−4.6 (3.1)	−4.3 (3.7)	−0.6 (0.2)	−0.2 (0.2)
1.00	50.0	51.2	−9.6 (0.3)	−7.7 (0.4)	−2.7 (1.4)	−3.3 (1.8)	−2.3 (2.9)	−0.1 (0.5)	1.3 (0.3)
Panel C: linear signal, performance fee before costs (bp p.a.)									
0.05		0.80	0.25	0.12	−0.55	−0.58	−0.10	−0.18	−0.03
0.15		0.98	0.68	0.62	−0.30	−0.10	0.12	−0.34	−0.02
0.40		1.62	1.32	1.29	−0.06	−0.18	0.58	−0.18	−0.10
1.00		2.24	2.02	1.98	0.27	0.49	1.32	−0.07	0.04
Panel D: nonlinear signal, performance fee before costs (bp p.a.)									
0.05		0.57	−0.36	−0.23	−0.65	−0.64	−0.38	−0.40	−0.22
0.15		0.87	−0.42	−0.30	−0.39	−0.33	−0.37	−0.60	−0.19
0.40		1.77	−0.59	−0.40	−0.56	−0.40	−0.43	−0.57	−0.21
1.00		2.85	−0.29	−0.18	0.13	−0.15	0.17	−0.31	−0.05

Monte Carlo evaluation of the learners on simulated excess returns. The predictors (five lags of the ON and 1W rates), the sample with its gaps and the backtest design are those of the ON-1W pair; the target is $y_t = \sqrt{\text{SNR}}\, s(x_t) + e_t$ with $\text{Var}(s) = \text{Var}(e) = 1$, so that the population R^2 of the true conditional mean is $\text{SNR}/(1+\text{SNR})$. Panel A: linear signal, the projection of the actual ON-1W excess return on the predictors. Panel B: purely nonlinear signal, the interaction of the 1W–ON spread with the change in the ON rate, orthogonalised on the predictors. The noise e_t is a moving average of five i.i.d. normal shocks, which reproduces the overlap of the excess returns. 'Oracle' is the R^2_{oos} of the true conditional mean. Entries are R^2_{oos} in percent relative to the prevailing mean, averaged over independent noise draws (standard deviation across draws in parentheses). Panels C and D: the simulated target is mapped into an excess return with the mean and volatility of the actual ON-1W excess return, and the entries are the performance fees of Table S15 before transaction costs of the oracle and of the learners, in basis points a year; with a bid–ask spread of 4 basis points, all fees, including those of the oracle, are negative.

Table S14: **Signal-to-Noise Simulation: Rejection Rates**

SNR	VAR (DST)	Ridge	Random forest	Boosting	Random features	NN (4-2)	NN (32-16-8-4-2)
Panel A: linear signal							
0.05	100	100	0	0	33	0	0
0.15	100	100	0	67	100	33	33
0.40	100	100	67	33	100	33	0
1.00	100	100	100	100	100	67	100
Panel B: nonlinear signal							
0.05	0	0	0	0	0	0	0
0.15	0	0	0	0	0	0	0
0.40	0	33	33	67	67	67	0
1.00	0	0	100	100	100	100	100

Share of the simulated samples (in percent) in which the Clark–West test rejects $H_0 : R^2_{oos} \leq 0$ at the 5% level, for the designs of Table S13.

S6 Economic Value and Robustness

Table S15 reports the performance fees and risk-adjusted abnormal returns of DST's four cases (with and without costs and position limits). Table S16 varies risk aversion and the cost convention. Table S17 collects the robustness checks: episodes and loss function, out-of-sample tests under the simulated EH, stability over time (Figure S3), design choices (Table S18), the holdout, and break-even spreads.

Table S15: **Economic Value of Departures from the EH**

	VAR (DST)		Ridge		Random forest		Boosting		NN ensemble		Combination		EWMA mean	
	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2
Panel A: Case 1: no transaction costs, unconstrained weights														
ON-1W	1.80	2.00	1.62	1.79	0.89	0.95	1.10	1.18	0.94	1.00	1.69	1.88	0.48	0.50
ON-2W	0.83	0.96	0.83	0.95	0.79	0.90	0.75	0.85	−0.86	−0.91	1.21	1.42	0.87	1.01
ON-1M	0.33	0.35	0.86	0.95	0.92	1.01	0.86	0.94	0.56	0.61	1.01	1.12	0.99	1.09
1W-2W	2.17	2.29	2.13	2.24	1.24	1.27	1.20	1.22	−0.03	−0.03	1.84	1.92	0.61	0.61
Panel B: Case 2: transaction costs, unconstrained weights														
ON-1W	−1.41	−1.41	−1.37	−1.37	−1.41	−1.41	−1.42	−1.41	−0.60	−0.72	−0.92	−1.00	0.34	0.34
ON-2W	−1.45	−1.57	−1.07	−1.19	−0.51	−0.64	−0.35	−0.43	−1.00	−1.09	−0.06	−0.10	0.81	0.93
ON-1M	−0.29	−0.33	0.52	0.56	0.34	0.36	0.23	0.23	0.45	0.49	0.49	0.53	0.83	0.88
1W-2W	−1.80	−1.61	−1.63	−1.46	−1.21	−1.14	−0.61	−0.59	−0.24	−0.24	−0.66	−0.65	0.38	0.37
Panel C: Case 3: transaction costs, $-1 \leq w_t \leq 2$														
ON-1W	−1.03	−1.05	−0.99	−1.01	−1.12	−1.13	−1.10	−1.11	−0.23	−0.23	−0.56	−0.59	0.21	0.30
ON-2W	−1.12	−1.29	−0.80	−0.97	−0.34	−0.52	−0.27	−0.40	−0.46	−0.66	0.05	−0.06	0.78	0.86
ON-1M	−0.27	−0.32	0.52	0.54	0.37	0.39	0.25	0.26	0.45	0.49	0.51	0.54	0.84	0.88
1W-2W	−1.27	−0.98	−1.11	−0.86	−0.89	−0.72	−0.39	−0.35	−0.23	−0.23	−0.50	−0.43	0.39	0.38
Panel D: Case 4: transaction costs, no short selling ($0 \leq w_t \leq 1$)														
ON-1W	−0.47	−0.47	−0.45	−0.44	−0.51	−0.52	−0.50	−0.52	−0.07	−0.07	−0.24	−0.24	0.08	0.09
ON-2W	−0.44	0.00	−0.31	0.00	−0.09	0.00	−0.09	0.00	−0.12	0.00	0.07	0.00	0.38	0.00
ON-1M	−0.13	−0.12	0.27	0.25	0.20	0.17	0.13	0.11	0.23	0.20	0.26	0.25	0.44	0.45
1W-2W	−0.49	0.00	−0.41	0.00	−0.32	0.00	−0.11	0.00	−0.11	0.00	−0.18	0.00	0.21	0.00

A mean-variance investor with quadratic utility and relative risk aversion of 5 holds the long repo and a position $w_t = (\bar{\sigma}_t/\widehat{\sigma}_t)\,\text{sign}(\widehat{\mu}_t)$ in the strategy of rolling the short repo, where $\widehat{\mu}_t$ is the forecast of the excess return, $\widehat{\sigma}_t$ its conditional volatility and $\bar{\sigma}_t$ the average conditional volatility over the estimation window, so that the median position is about one unit of notional (Della Corte et al., 2008, Section 5). A new position is opened every day and held for the life of the long repo, so that each position replaces the one opened n business days earlier; transaction costs are a bid–ask spread of 4 basis points in rate terms on this change of the position. $\mathcal{F}$ is the annualised performance fee the investor would pay to switch from the strategy implied by the expectations hypothesis (prevailing-mean forecast) to the strategy based on the model forecasts, and M^2 the risk-adjusted abnormal return of Modigliani and Modigliani (1997), both in basis points per year. Out-of-sample forecasts with the lags of the pair as predictors.

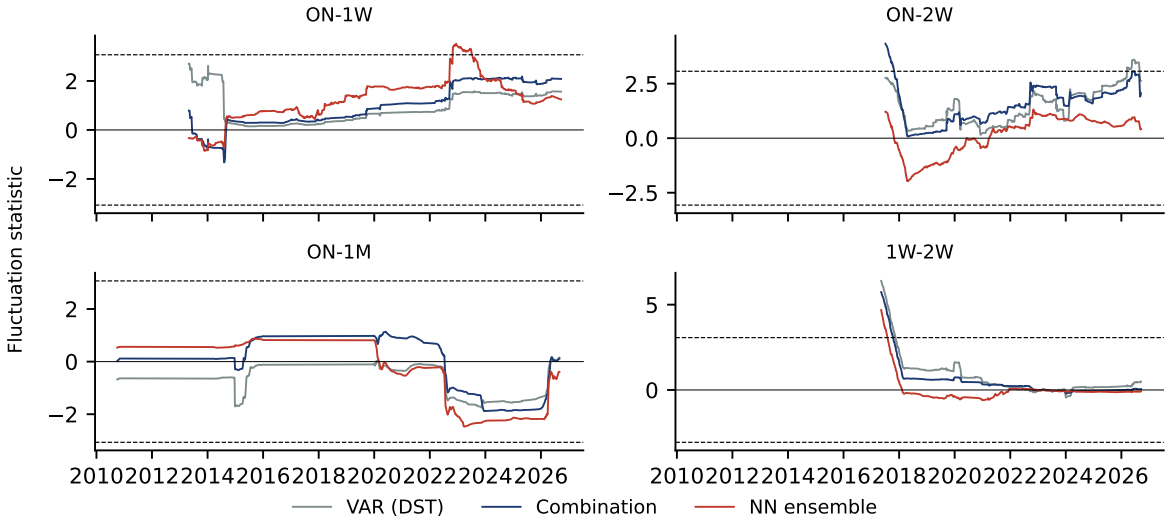


Figure S3: **Local Forecast Gains over Time.** Fluctuation statistic of Giacomini and Rossi (2010): standardised rolling mean of the difference between the squared errors of the EH forecast and those of the VAR, the combination of all forecasts, and the network ensemble, over a window of 30% of the out-of-sample period. Values above the upper dashed line indicate significantly smaller squared errors than the EH forecast at the 5% level.

Table S16: **Economic Value: Costs, Turnover, and Risk Aversion**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: Performance fee without costs (bp p.a.)				
VAR (DST)	1.80	0.83	0.33	2.17
Ridge	1.62	0.83	0.86	2.13
Random forest	0.89	0.79	0.92	1.24
Boosting	1.10	0.75	0.86	1.20
NN ensemble	0.94	−0.86	0.56	−0.03
Combination	1.69	1.21	1.01	1.84
EWMA mean	0.48	0.87	0.99	0.61
Panel B: Break-even bid–ask spread (bp)				
VAR (DST)	2.24	1.46	2.12	2.19
Ridge	2.17	1.74	10.18	2.26
Random forest	1.55	2.42	6.41	2.02
Boosting	1.75	2.74	5.44	2.65
NN ensemble	2.44	0.00	20.77	0.00
Combination	2.59	3.81	7.84	2.95
EWMA mean	13.55	56.25	24.57	10.54
Panel C: Average absolute change of the position per day				
VAR (DST)	0.830	0.593	0.233	1.035
Ridge	0.775	0.496	0.163	0.981
Random forest	0.604	0.347	0.222	0.654
Boosting	0.656	0.295	0.236	0.492
NN ensemble	0.413	0.056	0.105	0.092
Combination	0.681	0.338	0.206	0.665
EWMA mean	0.063	0.037	0.118	0.097
Panel D: Performance fee with costs, relative risk aversion 2 (bp p.a.)				
VAR (DST)	−1.41	−1.45	−0.29	−1.80
Ridge	−1.37	−1.07	0.52	−1.64
Random forest	−1.41	−0.52	0.34	−1.21
Boosting	−1.42	−0.35	0.22	−0.61
NN ensemble	−0.60	−1.00	0.45	−0.24
Combination	−0.92	−0.06	0.49	−0.66
EWMA mean	0.34	0.81	0.82	0.38
Panel E: Performance fee with costs, relative risk aversion 10 (bp p.a.)				
VAR (DST)	−1.41	−1.45	−0.29	−1.80
Ridge	−1.37	−1.07	0.52	−1.63
Random forest	−1.41	−0.51	0.35	−1.21
Boosting	−1.42	−0.34	0.23	−0.61
NN ensemble	−0.60	−1.01	0.45	−0.24
Combination	−0.92	−0.06	0.50	−0.65
EWMA mean	0.34	0.82	0.83	0.38
Panel F: Volatility of net returns relative to the EH strategy (bp p.a.)				
VAR (DST)	1.70	1.86	2.51	1.92
NN ensemble	1.23	2.24	1.88	0.27
Combination	1.53	1.55	1.97	1.61
EWMA mean	1.10	0.74	2.27	0.71
Panel G: Maximum drawdown relative to the EH strategy (bp)				
VAR (DST)	27.11	14.02	11.08	34.67
NN ensemble	12.95	14.17	2.80	3.04
Combination	21.27	7.54	2.86	21.30
EWMA mean	5.07	0.84	5.68	3.57

Economic value of the forecasts under alternative assumptions, without position limits. The break-even spread is the bid–ask spread in rate terms at which the performance fee falls to zero. Turnover is the average absolute change of the weight on the rolling strategy from one origin to the next, in units of notional. The baseline of Table S15 uses a spread of 4 basis points and relative risk aversion of 5. Panels F and G: returns of the strategy net of the bid–ask spread minus those of the EH strategy, measured over non-overlapping investment horizons. Lags of the pair as predictors.

Table S17: **Robustness**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: R^2_{oos} of the VAR (%), episodes and loss function				
Full sample	38.68***	25.40***	−17.26	49.55***
Without Sep 2008–Jun 2009	32.90***	25.40***	−23.74	49.55***
Without Sep–Dec 2019	38.67***	24.24***	−17.78	50.57***
Without Mar–Jun 2020	39.79***	31.83***	−19.29	54.81***
Without year-end windows	40.27***	23.16***	−16.58	48.13***
Without all four	36.27***	30.66***	−27.52	57.17***
Largest 1% of EH errors removed	24.56***	19.35***	−39.63	46.07***
Absolute-error loss	25.46***	15.27***	−15.37	25.56***
Panel B: Simulation under the EH (200 samples)				
VAR R^2_{oos}: 95th percentile under the EH (%)	−2.18	−1.58	−4.53	−1.30
VAR R^2_{oos}: bootstrap p-value	0.000	0.000	0.745	0.000
Clark–West: rejection rate at 5% (%)	3.5	7.5	7.5	20.5
Panel C: Share of time with significant local gains over the EH (%)				
VAR (DST)	0.0	5.0	0.0	5.6
Combination	0.0	3.5	0.0	4.6
Panel D: Design choices (9 designs), VAR				
Lowest R^2_{oos} (%)	35.17	17.54	−17.26	37.86
Highest R^2_{oos} (%)	41.10	26.01	−0.33	51.63
Designs with Clark–West $p < 0.05$	9/9	9/9	2/9	9/9
Panel E: Holdout, 21 April 2021 to the end of the sample, R^2_{oos} (%)				
VAR (DST)	62.01***	41.07***	−47.20	11.97***
Combination	43.12***	26.30***	−40.36	3.19***
NN ensemble	4.16*	4.39***	−20.98	−0.17
Panel F: Break-even bid–ask spread (bp)				
VAR (DST)	2.24	1.46	2.12	2.19
Combination	2.59	3.81	7.84	2.95

Panel A: R^2_{oos} of the VAR relative to the EH forecast after removing forecast origins whose investment window overlaps the stated periods, after removing the 1% of origins with the largest absolute errors of the EH forecast, and with absolute instead of squared errors; stars refer to the Clark–West test and, for absolute errors, to a one-sided Diebold–Mariano test. Panel B: rates are simulated from the two-factor model of Table S5, in which the EH holds exactly (for the out-of-sample test with expected short rates weighted by calendar days, as in the excess return), with shocks drawn by a wild bootstrap and the stale-quote pattern of the data; on each sample the out-of-sample VAR forecasts are computed exactly as in the data. Rejection rates at the nominal 5% level measure the size of the tests; bootstrap p-values are the shares of simulated statistics at least as large as in the data. Panel C: share of dates at which the fluctuation test of Giacomini and Rossi (2010), computed over a rolling 30% of the out-of-sample period, indicates significantly smaller squared errors than the EH forecast at the 5% level. Panel D: range of the VAR's R^2_{oos} and number of designs with a significant Clark–West statistic across the designs in Table S18. Panel F: bid–ask spread in rate terms at which the performance fee of Table S15 (no position limits) falls to zero; 0 if the fee is not positive without costs. Panel E: out-of-sample R^2 from 21 April 2021, a period whose data were not used to design the study (models, tests, and data cleaning were fixed on data until April 2021); '–': fewer than 30 forecasts.

Table S18: **Out-of-Sample Predictability under Alternative Designs**

	ON-1W	ON-2W	ON-1M	1W-2W
VAR (DST)				
Baseline (1,250 days)	38.68***	25.40***	−17.26	49.55***
Window 750 days	36.79***	17.54***	−10.18**	47.76***
Window 2,500 days	39.55***	22.11***	−10.86*	44.32***
Expanding window	41.10***	23.28***	−0.33**	37.86***
Validation share 25%	38.68***	25.40***	−17.26	49.55***
Re-estimation every 5 days	38.93***	26.01***	−14.31	49.48***
Stale threshold 10 days	37.58***	25.07***	−10.16	46.12***
Stale threshold 40 days	35.17***	25.40***	−16.75	49.55***
Stale quotes masked in real time	35.29***	25.06***	−15.20	51.63***
Ridge				
Baseline (1,250 days)	21.32***	22.33***	−18.01	48.25***
Window 750 days	17.06***	15.10***	−15.13	46.94***
Window 2,500 days	23.33***	19.41***	−13.77	44.74***
Expanding window	26.78***	22.07***	−5.74**	39.42***
Validation share 25%	22.93***	23.24***	−17.19	47.22***
Re-estimation every 5 days	21.73***	22.39***	−15.26	47.88***
Stale threshold 10 days	26.78***	23.50***	−12.59	44.81***
Stale threshold 40 days	27.70***	22.33***	−13.59	48.25***
Stale quotes masked in real time	27.63***	24.50***	−10.70	50.38***
Lasso				
Baseline (1,250 days)	21.74***	12.38***	−22.11	41.76***
Window 750 days	16.81***	7.48***	−22.60	42.43***
Window 2,500 days	22.80***	13.81***	−11.08	44.24***
Expanding window	27.35***	18.37***	−6.06**	38.14***
Validation share 25%	21.29***	14.83***	−27.26	45.81***
Re-estimation every 5 days	21.86***	12.67***	−19.43	40.99***
Stale threshold 10 days	19.06***	14.00***	−15.52	44.73***
Stale threshold 40 days	35.20***	12.38***	−12.82	41.76***
Stale quotes masked in real time	36.65***	13.83***	−10.11	44.20***
Random forest				
Baseline (1,250 days)	−54.38	−1.43***	−56.43*	27.34***
Window 750 days	−69.06	−15.33***	−60.33**	32.86***
Window 2,500 days	−55.18	8.91***	−28.82*	−46.34***
Expanding window	−46.39	13.14***	−26.10*	7.82***
Validation share 25%	−54.13	−2.11***	−56.07*	30.62***
Re-estimation every 5 days	−33.87	4.10***	−47.77*	32.78***
Stale threshold 10 days	−63.81	−3.07***	−54.69**	28.62***
Stale threshold 40 days	−24.35**	−1.43***	−37.09*	27.34***
Stale quotes masked in real time	−16.13***	6.33***	−31.09**	30.42***
Boosting				
Baseline (1,250 days)	−72.97*	6.24***	−34.37*	26.52***
Window 750 days	−128.92	−8.03***	−45.17**	29.28***
Window 2,500 days	−90.06*	−5.16***	−6.57**	17.52***
Expanding window	−48.35*	7.39***	−8.17**	22.52***
Validation share 25%	−75.32*	−18.80***	−56.57*	30.75***
Re-estimation every 5 days	−41.03**	0.44***	−25.41	24.23***
Stale threshold 10 days	−119.75	−31.35***	−46.78**	24.02***
Stale threshold 40 days	−96.82	6.24***	−36.10*	26.52***
Stale quotes masked in real time	−39.72	−4.95***	−35.97*	29.59***
Sup test across windows and validation shares, p-value				
VAR (DST)	0.003	0.000	0.077	0.000
Ridge	0.000	0.000	0.116	0.001
Lasso	0.000	0.000	0.211	0.001
Random forest	0.475	0.000	0.066	0.010
Boosting	0.146	0.000	0.034	0.011

R^2_{oos} (in percent) relative to the EH forecast under alternative designs, with Clark–West significance stars; each design is evaluated on its own out-of-sample period. The stale-quote thresholds change the evaluation sample and the network ensemble comprises the architectures estimated under each design. The sup test uses the largest Clark–West statistic across the designs that share the evaluation sample (window lengths and validation share), on their common dates, with a critical value from a joint stationary bootstrap, and so accounts for the search over designs (in the spirit of Rossi and Inoue, 2012). Lags of the pair as predictors.

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